XINGQIANG CHEN

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OBJECTIVE

Looking for **FULL TIME** roles in **AI Accelerator Compiler System Software LLM Inference Systems HW/SW Co-design**.

INTERDISCIPLINARY BACKGROUND

Physics and biophysics training across computational chemistry, molecular dynamics, and computer science. Contests: **CUMCM National First Prize** (2012); **MCM Meritorious Winner** (2013).

EDUCATION

Ph.D. in Biophysics and Condensed Matter Sep. 2014 - Jul. 2018
Xiamen University, Department of Physics and Engineering GPA : 3.5/4.0

B.S. in Theoretical Physics Sep. 2010 - Jul. 2014
Xiamen University, Department of Physics and Engineering GPA : 3.0/4.0

SKILLS AND PROJECTS

Programming Languages: C++, Python; Java, Scala, Go, CUDA.

MLOps Tools: PyTorch, TensorFlow, Triton, TVM, Kubernetes; Hadoop, Flink, Spark.

LLM Inference Systems: Transformers, KV-Cache, quantization, batching & scheduling, Triton kernels, CUDA, FlashAttention, graph fusion, GPU profiling & replay.

AI Compiler Kernel: MLIR graph compiler, CIM/NPU codegen, heterogeneous runtime; **Palladium** emulation and HW/SW co-debug; YiRage YPK; TraceSmith.

Deep Learning Models: Knowledgeable in Transformers, RL, GNN, DNN, CNN, and more.

Interdisciplinary Expertise: Background in Physics, Computational Biology, Computational Chemistry, Molecular Dynamics, Quantum Mechanics simulations, and Computer Science.

Open Source Highlights:

- **TraceSmith** — Cross-platform GPU Profiling & Replay System for AI compilers and DL frameworks (CUDA/ROCm/Metal/M lock-free capture, deterministic replay; PyPI). `https://github.com/chenxingqiang/TraceSmith`
- **Research CLI** — Command-line AI workflow tool with academic research features; multi-model providers and MCP tooling. `https://github.com/iechor-research/research-cli`
- **JAX-sklearn** — Drop-in scikit-learn replacement with automatic JAX acceleration (*13,058 tests passed; PyPI release*). `https://github.com/chenxingqiang/jax-sklearn`
- **YouTube-DNN-RecSys** — Clean TensorFlow & Spark implementations for YouTube-style recommendation models (Python+Scala). `https://github.com/chenxingqiang/YouTube-DNN-RecSys`
- **Questions-Gen** — Math competition problem generator (Qwen3-14B, RL GRPO, KD; CLI & HF models). `https://github.com/chenxingqiang/GenQuestion`
- **DeepMicroPath** — Multi-mode research agent with microbiology specialization; non-intrusive wrapper enabling auto/default/microbiology prompts; Docker/SLURM deployment. `https://github.com/chenxingqiang/DeepMicroPath`
- **Medical-NER-Qwen-4B-Thinking** — Qwen3-4B-Thinking fine-tune via LoRA for medical entity & relation extraction (English). `https://huggingface.co/xingqiang/Medical-NER-Qwen-4B-Thinking`
- **QuestionsGen-Qwen3-14B-Final (FP)** — Full-precision merged final model trained with Unsloth; deployment-ready HF weights. `https://huggingface.co/xingqiang/QuestionsGen-Qwen3-14b-final-fp-merged`
- **YiRage Persistent Kernel (YPK)** — LLM megakernel compiler & runtime achieving 1.2×–6.7× latency reduction. `https://github.com/chenxingqiang/YiRage`

EXPERIENCE

Product-Line Compiler Lead, Witmem ( ) — Hangzhou, China
Product-line Compiler / Palladium Emulation — *Dec. 10, 2025 - Present*

- Own a CIM product line's **compiler stack** and **Palladium** hardware emulation: close the loop from workload → graph compile/codegen → on-chip execution → emulation → root-cause, tracing software-stack bugs to hardware behavior.
- Lead software bring-up on Palladium: run compiler-generated programs on the emulator, HW/SW co-debug (GDB + model/silicon/system software), and close functional/perf issues—not just mechanical regression.
- Built **CIM-MLIR**, an **AI graph compiler** (not ISA/LLVM-backend only): PyTorch/TensorFlow/ONNX via TOSA/StableHLO onto CPU/GPU/CIM, filling the product line's missing AI-compiler capability.
- Cost-driven min-cut partition plus CIMHigh/Mid/Low IR: GEMM-like ops on CIM, nonlinear on GPU, control on CPU; OR-Tools Die/L2/L0 allocation; emit burnable **config.bin/map.bin** plus PTX and LLVM host code.
- Shipped the Runtime/Firmware stack: per-device executors, unified memory, compute/transfer overlap and triple buffering; C++/Python APIs for latency, utilization, and transfer cost on LLM/CNN CIM deployments.
- Research from this role: **GRC-PIM**, a drift-gated, GEMV-reusing residual calibrator for soft-decision error correction in PIM; submitted to *npj Unconventional Computing* (Future Computing collection).

Senior AI Compiler Engineer, Yizhu Tech Ltd. — Hangzhou, China
R&D Department — *Sep. 2022 - Dec. 2025*

- Led development of **YiRage Persistent Kernel (YPK)**, compiling end-to-end LLM inference into a single megakernel; on Llama2-7B achieved **1.2×–6.7×** latency reduction, **40%** higher GPU utilization, and **25%** lower memory use.
- Designed fused operators (RMSNorm+Linear, etc.), cutting **85%** of kernel-launch overhead; single-request latency from **120ms to 18ms** (batch size=1).
- Built adaptive scheduling with Triton and FlashInfer; scaled to **8 GPUs** at **15,000 tokens/s**.
- Developed a graph compiler with kernel auto-tuning and KV-cache layout; memory-access efficiency **+60%**, cache hit rate **92%**.
- Filed **2** patents; deployed to **2** datacenter customers.
- Owned ReRAM SoC compiler backend and software ecosystem: LLVM Target, ISA bring-up, and code standardization.
- Led model conversion and operator-level analysis (Cycles/MACs) across runtimes; integrated TVM, ONNX, MLCommons, TensorFlow/PyTorch into YZlite for chip benchmarks.

Privacy-preserving Algorithms Engineer, Trustbe ( ) — Hangzhou, China
IT Group — *Sep. 2021 - Sep. 2022*

- Built a homomorphic-encryption federated learning platform for **20+** financial institutions; **15%** higher accuracy with privacy preserved, **40%** shorter training time.
- Designed MPC protocols for joint risk models at **8** banks; fraud detection **96.7%** accuracy, **30%** lower false positives.
- Open source: **Secret-Learn**, **573** implementations (191 algorithms × FL/SS/SL), 100% sklearn-compatible, JAX **5×+** speedup, SecretFlow integration (**PyPI, 150,000+ LOC**).
  `https://github.com/chenxingqiang/Secret-Learn`
- Published **2** papers and filed **3** patents; core tech adopted by **5** products.

Senior Algorithms Engineer, Yalla Group (NYSE: YALA) — Hangzhou, China
Data Center Engineering Group — *Sep. 2020 - Sep. 2021*

- Rebuilt the ChatRooms recommender with multi-objective optimization; daily active time **+35%**, 7-day retention **42% → 61%**.
- Built Ludo risk control with real-time behavior models; malicious-user detection **94%**, refund rate **−45%**.
- Developed a Texas Hold'em game AI with deep RL; **78%** win rate, play time **+25%**, paid conversion **+18%**.

Senior Algorithms Engineer, Gejia Network Tech — Hangzhou, China
Search & Recommendation Engineering Group — *Mar. 2019 - Sep. 2020*

- Designed multi-scenario ranking with CF, deep models, and knowledge graphs; CTR **+28%**, CVR **+35%**.
- Optimized the real-time recommender to **1M** QPS with **<50ms** latency, **95%** coverage, satisfaction **3.2 → 4.6**.

Machine Learning Engineer, Bianlifeng Tech — Beijing, China
Data Engineering Group — *Mar. 2017 - Mar. 2019*

- Built sales forecasting for **50+** stores; **92%** accuracy, inventory turnover **+30%**.
- Linked customer IDs to purchases over **10M+** events; identity accuracy **89%**, personalized sales **+22%**.

- Shipped a CV checkout model for **500+** SKUs; **96%** accuracy, checkout speed **+60%**.

RESEARCH & INTERNSHIPS

Molecular Dynamics Simulation of Proteins with QM/MM model
Xiamen university, China
(China National Funds) *Sep. 2014 - Mar. 2017*

- Conducted research on protein dynamics simulations using quantum mechanics, and analyzed biological information of proteins. This involved interdisciplinary knowledge in theoretical physics, structural biology, computational chemistry, and computer science.

Development of Liquid Biopsy ct-DNA Kit and DNA Methylation Data Analysis
Lepu Medical, China
(R&D Engineer Intern) *Mar. 2016 - Mar. 2017*

- Worked on the development of liquid biopsy kits, which required specific biomarkers for cancer screening. Using existing gene chip data (TCGA&GEO), we analyzed 10,000 samples to locate the methylation site on the Septin9 human gene, an early indicator of colon cancer. This informed the development of a more accurate clinical diagnostic kit for colon cancer.

AWARDS

- Judge, First Screening on Best Practices of Applied Algorithms, 2021.
- Ranks 57 and 32 out of 1739, IEEE ICDM Global Meteorological AI Challenge, 2018.
- Meritorious Winner, Mathematical Contest in Modeling, North American College Students, 2013.
- 6th place, International Solar Energy Decathlon Competition, 2013.
- Outstanding Prize, Contemporary Undergraduate Mathematical Contest in Modeling, 2012.

PUBLICATIONS

- Chen X. GRC-PIM: A Drift-Gated, GEMV-Reusing Residual Calibrator for Soft-Decision Error Correction in Processing-in-Memory. *npj Unconventional Computing* (Future Computing collection, submitted; work at Witmem /).
- Cai Y, Su M, Ahmad A, Hu X, Sang J, Kong L, Chen X, Wang C, Shuai J, Han A. Conformational dynamics of the essential sensor histidine kinase WalK. *Acta Crystallographica Section D: Structural Biology.* 2017 Oct 1;73(10):793-803.
- Xia H, Chen X, Chen X, Cheng J, Liu Y, Chen X, Zhang Q, Zou G. Chiral discrimination of amino acid enantiomers based on different interactions with Cu2+. *Sensors and Actuators B: Chemical.* 2018 Jan 1;254:44-51.
- Ahmad A, Cai Y, Chen X, Shuai J, Han A. Conformational dynamics of response regulator RegX3 from Mycobacterium tuberculosis. *PloS one.* 2015 Jul 22;10(7):e0133389.
- Chen Y, Qi H, Li X, Cai M, Chen X, Liu W, Shuai J. Suppressing effect of C a 2+ blips on puff amplitudes by inhibiting channels to prevent recovery. *Physical Review E.* 2016 Aug 15;94(2):022411.